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Department of Computer Engineering
CEN 302-Software Engineering**

PMS (Police Management System)

Requirements Specification

Version 1.0

March 26th, 2026

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1. Executive Summary

1.1 Project Overview

The Police Management System (PMS) is a comprehensive, web-based digital platform designed to modernize and streamline law enforcement operations within police departments. Traditional police record-keeping and case management often rely on paper-based documentation, disconnected databases, and fragmented communication channels between officers, detectives, supervisors, and administrators. These inefficiencies can lead to delayed investigations, data inconsistencies, loss of critical evidence records, and reduced accountability across departments.

The PMS addresses these challenges by providing a centralized, role-based digital platform that integrates incident reporting, case management, evidence tracking, suspect assignment, officer coordination, and administrative oversight into a single unified system. The platform ensures that each participant in law enforcement operations acts within clearly defined responsibilities, improving accountability, data integrity, and operational efficiency.

The system introduces a multi-role architecture consisting of Police Officers, Detectives, Supervisors, and Administrators. Each role interacts with the system through a tailored interface that reflects the specific permissions and responsibilities associated with their position, enabling efficient collaboration while maintaining strict access control and data security.

Key features of the Police Management System include:

- Secure user authentication and role-based access control
- Incident report creation and management by authorized police officers
- Case management including suspect assignment, evidence tracking, and status updates
- Structured inspection and reporting workflows for detectives
- Supervisor-level case assignment and officer coordination
- A real-time administrative dashboard with statistics and system monitoring
- Comprehensive user and role management by administrators

By digitizing and automating these workflows, the platform enhances transparency, improves inter-departmental communication, and ensures that all case-related data is stored securely in a centralized and accessible database.

Accessible from any device with internet connectivity, the PMS offers a user-friendly and responsive interface that supports seamless interaction between all user roles. The system ultimately aims to increase investigative efficiency, reduce manual overhead, and provide a reliable, secure, and accountable platform for law enforcement management.

1.2 Purpose and Scope of this Specification

Purpose

This specification document defines the functional and non-functional requirements of the Police Management System. It serves as a comprehensive guide for developers, stakeholders, and evaluators to understand the system's architecture, features, and operational behavior.

The purpose of this project is to design and develop a robust law enforcement management platform that digitalizes and automates police operations, from incident report creation to case closure. The platform facilitates structured collaboration between multiple user roles while ensuring data integrity, security, and transparency throughout all operational processes.

Additionally, this document aims to establish a clear understanding of system functionalities, including user role permissions, workflow processes, and interaction mechanisms. It also supports effective communication within the development team, ensuring alignment with project goals and defined requirements.

Scope

In Scope

The Police Management System includes the following core functionalities:

- **User Authentication and Access Control:** Secure login for all users with role-based permission enforcement throughout the system.
- **Incident Report Management:** Enables Police Officers to create, document, and submit incident reports for recorded crimes.
- **Case Management:** Supports creation, updating, and status tracking of cases (Open, Under Investigation, Closed) by authorized personnel.
- **Suspect and Evidence Management:** Allows Detectives and Officers to link suspects and physical/digital evidence to existing cases.
- **Officer Assignment:** Enables Supervisors to assign officers to specific cases, ensuring proper investigative coverage.
- **User and Role Administration:** Provides Administrators with full control over user accounts, roles, and department management.
- **Statistics Dashboard:** Displays real-time data including number of cases, officers, incidents, and department performance metrics.
- **Arrest Record Management:** Allows Officers to document arrest information and link it to relevant case files.

Out of Scope

The following aspects are not included in the current scope of the Police Management System:

- Integration with external national or international law enforcement databases (e.g., Interpol, national criminal registries).
- Real-time GPS tracking or surveillance integration for patrol officers.
- Automated court case management or legal documentation generation.
- Mobile-native applications; the system is web-based only, though the interface is responsive.
- Biometric authentication systems such as fingerprint or facial recognition.
- Post-deployment maintenance and support services, which will be defined separately.

2. Product / Service Description

2.1 Product Context

The Police Management System is a standalone, self-contained web-based platform that does not depend on or integrate with any external law enforcement systems or third-party databases in its current version. All core workflows — incident report creation, case management, evidence tracking, suspect assignment, officer coordination, and administrative oversight — are handled entirely within the platform.

The system interfaces internally between four distinct user roles: Police Officers, Detectives, Supervisors, and Administrators. Each role interacts through a dedicated, permission-restricted interface within the same environment, ensuring that no user can access or perform actions outside their designated responsibilities.

The PMS operates under a client-server architecture, where all users interact with the application through a standard web browser, while data is processed and stored on a central server. This ensures real-time updates, consistent data visibility, and synchronized operations across all roles.

While the platform operates independently in its initial version, it relies on a hosting infrastructure for deployment and availability, and on an email/notification service to deliver real-time alerts to users upon relevant system events such as case assignments, status changes, or report submissions.

2.2 User Characteristics

Police Officer

Represents the frontline law enforcement personnel responsible for documenting criminal incidents and managing field-level case records.

- Registers and logs into the system using department-issued credentials managed by the Administrator.
- Creates new incident reports for reported crimes, entering key details such as location, crime type, description, date, and time.
- Updates case information and status (Open, Under Investigation, Closed) as investigations develop.
- Uploads and attaches physical or digital evidence to existing cases.
- Records and documents arrest information linked to relevant cases.
- Views all cases assigned to them from a personalized dashboard for efficient investigation management.
- Cannot manage user accounts, modify other officers' reports, or access administrative settings.
- Requires basic-to-moderate familiarity with web-based platforms and standard digital record-keeping tools.

Detective

Represents specialized investigative personnel responsible for managing complex cases, coordinating with officers, and maintaining detailed investigative records.

- Logs into the system using credentials issued by the Administrator.
- Views all cases assigned to them and accesses full case details including incident reports and evidence.
- Assigns and links suspects to existing cases, entering details such as name, age, physical description, and associated evidence.
- Uploads and manages evidence related to ongoing cases in coordination with police officers.
- Reviews and analyzes all information associated with a case before determining investigative direction.
- Cannot create new incident reports, manage user accounts, or perform administrative actions.
- Requires professional investigative knowledge, strong analytical skills, and proficiency with digital case management tools.

Administrator

Oversees the overall integrity of the platform, managing user accounts, roles, and department structure, while also coordinating officer assignments and monitoring investigative activity across the system. The Administrator serves as the highest-authority role and has visibility over all system operations.

- Pre-configured in the system at deployment time; no self-registration or signup flow exists for this role.
- Creates new user accounts for Police Officers and Detectives, providing them with login credentials and appropriate role assignments.
- Updates, deactivates, or reactivates existing user accounts as needed, notifying affected users with documented reasons.
- Assigns and modifies user roles to ensure each person has the correct level of access and permissions within the system.
- Adds, edits, and manages police department records so that officers can be organized by their respective unit or department.
- Assigns Police Officers to specific cases, ensuring investigations are handled by the appropriate personnel.
- Monitors all active and ongoing cases within the system, tracking case statuses and officer workloads.
- Monitors system-wide activity through an administrative dashboard, reviewing user actions and ensuring platform integrity.
- Views real-time analytics including number of cases, registered users per role, active incidents, and department performance metrics.
- Cannot create incident reports, upload evidence, or assign suspects to cases, as these actions belong to Officers and Detectives.
- Requires strong administrative and organizational judgment, familiarity with law enforcement workflows, and proficiency with dashboard-based monitoring tools.

2.3 Assumptions

- All users are assumed to have basic-to-moderate familiarity with web-based platforms and standard digital tools.
- The system operates under the assumption of a stable and reliable internet connection for all users.
- Police Officers are responsible for providing accurate and complete information when creating incident reports.
- Detectives are assumed to act in good faith when linking suspects and evidence to cases within the boundaries of their role.
- Administrators will complete all pending user verification and account management tasks within a reasonable timeframe.
- All uploaded evidence and documentation is assumed to be authentic and accurately represents the events described.
- All users are responsible for maintaining the confidentiality of their login credentials and not sharing access.
- Notifications and alerts sent by the system are assumed to reach users in a timely manner through the platform's notification mechanism.
- The system assumes that user roles are assigned accurately by the Administrator and reflect actual departmental positions.
- It is assumed that the system will perform regular data backups to prevent data loss in the event of system failure or unexpected issues.
- Users are assumed to follow departmental procedures and legal standards when entering case-related data into the system.

2.4 Constraints

- The platform is web-based and requires a stable internet connection for all operations; offline functionality is not supported.
- The system must comply with applicable data protection and law enforcement data regulations to ensure the privacy, security, and legal integrity of all stored information.
- Role-based access control is strictly enforced; users cannot perform actions outside the permissions defined for their assigned role.
- User accounts for Officers, Detectives, and Supervisors can only be created by an Administrator; no self-registration flow is available for these roles.
- Incident reports, once formally submitted and saved, should follow a controlled editing process to maintain data integrity.
- The system does not support integration with external national law enforcement databases or third-party systems in its current scope.
- The platform does not support legal contract generation, court case management, or automated digital signing of legal documents.
- The user interface and functional behavior may be constrained by the limitations of the web technologies and frameworks used during development.
- Budget and timeline constraints during development may limit the implementation of certain advanced features or performance optimizations.
- The system is not a mobile-native application; while the interface is designed to be responsive, it is delivered and optimized as a web-based platform.

2.5 Dependencies

- The activation and availability of all Officer, Detective, and Supervisor accounts is dependent on the Administrator creating and configuring them correctly.
- The creation of a case is dependent on an Incident Report being submitted first by an authorized Police Officer.
- The assignment of a suspect to a case is dependent on the existence of an active case within the system.
- The upload of evidence is dependent on a case already being created and accessible within the system.
- The assignment of Officers to specific cases is dependent on the Supervisor having visibility into active cases and available personnel.
- The generation of dashboard statistics is dependent on accurate and complete data being entered by all users across all roles.
- Notifications and system alerts are dependent on the platform's notification service operating correctly and user email addresses being properly configured.
- The dispute or access issue resolution process is dependent on the Administrator having full visibility into all user accounts, case records, and system logs.
- The platform's overall performance and availability are dependent on the reliability and capacity of the chosen hosting infrastructure.
- The successful development of the project is dependent on the use of specific technologies for backend, frontend, and database management as defined in the technical architecture.